Mammalian Genome 7, 47–51 (1996).

Mammalian Genome

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Accurate mapping of the "acid meat" *RN* gene on genetic and physical maps of pig Chromosome 15

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Received: 25 April 1995 / Accepted:

Abstract. It has been shown that a major gene, called *RN*, is responsible for the RTN technological yield, a meat quality porcine trait. Experimental families informative for the segregation of *RN* gene were constituted from animals belonging to the Laconie composite line. We have previously mapped the *RN* gene to Chromosome (Chr) 15 (Milan et al. Genet. Sel. Evol. 27, 195–199, 1995). A Chr 15 map was established with 16 markers. The *RN* gene was found to be located between markers *Sw120* and *Sw936*, at 2 cM from *Sw936* (LOD = 38.1). In addition, by localizing *Sw936* at 15q21–22 using DISC-PCR, we also located *RN* on the physical map.

Introduction

The Hampshire pigs have a higher intramuscular concentration of glycogen than other breeds (Monin and Sellier 1985). After slaughtering, glycogenolysis in the muscles leads to meat acidification. To study the determinism of acid meat, Naveau analyzed the so-called "Napole" technological yield (RTN), corresponding to the meat yield after cooking of a semi-membranous muscle sample (Naveau 1986). In animals from Laconie synthetic line, founded by Hampshire, Piétrain, and Large White breeds in equal proportion, he identified the involvement of a major gene called *RN* (Naveau 1986; Le Roy et al. 1990). The effect of the *RN* gene on RTN was estimated to be about 3 phenotypic standard deviations between the means of homozygotes, with an unfavorable RN^- allele fully dominant over the rn^+ allele. Experimental families were produced to confirm the existence of the *RN* major gene (Le Roy et al. 1994), to estimate its effects on production traits (Le Roy et al. 1995) and to search for marker genes of the *RN* locus (Milan et al. 1995).

Meanwhile, over 500 microsatellite markers have been used to establish genetic maps of the porcine genome (Archibald et al. 1995; Ellegren et al. 1994; Rohrer et al. 1994). A set of markers covering the genome could thus be defined and used to map the *RN* gene on pig Chr 15 (Milan et al. 1995). The relatively large number of microsatellite markers available on this chromosome allowed us to develop a map of Chr 15 on our experimental population and to provide a more accurate mapping of the *RN* gene.

Materials and methods

Experimental families. The production of experimental families has been described elsewhere (Le Roy et al. 1994; Milan et al. 1995). Briefly, animals from matings between heterozygous parents were progeny tested, with rn^+/rn^+ mates, to determine their RN genotypes. Four males and nine females were identified as heterozygotes with a probability greater than 0.96. Families from these 13 animals correspond to backcross pedigrees regarding the *RN* gene. Segregation analysis of the RTN data in these families allowed the selection of 204 offspring for which the *RN* genotype, that is, the allele coming from their heterozygous parent, is known with an estimated probability greater than 0.9. Thus, 60 male informative meioses and 144 female informative meioses, with phase unknown, were available for mapping the *RN* gene.

PCR amplification of previously published markers of Chr 15. Primers for published Chr 15 markers were synthesized. We also synthesized primers for markers *Sw1119* and *Sw604*, using information available on WWW server of MARC (http://sol.marc.usda.gov/genome/swine/htmls/pig15.html). The unpublished sequences of primers for *Sw604* were kindly communicated by Craig Beattie. For each set, one of the primers was labeled with either 6-Fam, Hex, or Tamra fluorescent dye. All primers were purified. PCR reactions were carried in 25 µl containing 50 ng of DNA, 200 µM dNTP, 0.5 µM of primers, 0.5 U of Taq polymerase (GIBCO), 1 × W1 buffer (GIBCO); $MgCl_2$ concentrations and annealing temperature were as described by the authors. After 5 min at 94°C, 30 cycles of 45 s at the specific annealing temperature and 45 s at 94°C were performed. A last cycle with 10 min of elongation at 72°C was added. Markers *Sw120* and *Sw604* were efficiently amplified only by Touch Down PCR (Don et al. 1991). The annealing temperature was decreased by 1°C every two cycles between 64°C and 60°C during the 10 first cycles. Twenty-five standard cycles were subsequently performed with an annealing temperature of 59°C.

Development of two new markers of Chr 15. New markers have been developed as described previously (Robic et al. 1994), of which two (*S0355* and *S0369*) were mapped to Chr 15 in the PiGMaP reference families. The primers for *S0355* are: 5′-TCTGGCTCCTACACTCCTTCT-TGATG-3′ and 5′-TTGGGTGGGTGCTGAAAAATAGGA-3′, and for *S0369* are 5′-GAGAGGAAGGAGAAGGAAACAAG-3′ and 5′-AGGTCTATGTGTCAGCATAAAGAA-3′. The experimental conditions to use for these markers are listed in Table 1.

Genotyping on ABI sequencer. After PCR, the amplified products analyzed in the same gel lane were pooled before precipitation. The pellet was resuspended in 2.5 µl of formamide and 0.5 µl of Genescan 350. The samples were denatured at 95°C for 2 min and kept on ice before loading. The samples were loaded on 36 lanes of 12-cm gels (6% acrylamide/8.3 M urea), in an ABI 373 automatic sequencer. Electrophoresis was carried out for 3H30 at 800 V. After electrophoresis, the sizes of all fragments were analyzed by Genescan software. Genotypes were then interpreted, verified, and stored with GEMMA software (Milan and Woloszyn 1994). GEMMA contains procedures that facilitate the various steps from the DNA management to the calculation of two-point genetic distances.

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Table 1. Panel of markers of Chr 15.

Marker	Alleles[a]		PIC[b]	No. Inf. Meioses[c]	Ref.[d]	[Mg Cl_2] (mM)	Temp. (°C)	Quality[e]
	Described	Found						
S0355	-	3	0.40	165	A	4	55	**
Sw919	11	6	0.66	-	B	1.5	58	*
Sw1111	6	5	0.7	285	B	1.5	65	***
Sw964	5	6	0.73	255	B	1.5	58	*
Swr283	3	4	0.57	372	B	1.5	60	***
S0369	-	3	0.43	223	A	3	55	*
S0088	5	2	0.34	218	C, D, B	1.5	58	***
Sw1065	3	2	0.36	164	B	1.5	60	***
S0149	6	6	0.71	282	E, B	1.5	50	***
Sw1263	5	3	0.53	379	B	3	60	*
Sw423	5	3	0.30	71	B	1.5	60	**
Swr1002	4	3	0.51	238	B	2	58	***
Sw120	6	4	0.56	234	B	4	64–59[f]	*
Sw936	6	2	0.34	161	B	1.5	58	***
Sw906	7	2	0.13	39	B	1.5	55	*
Sw604	5	3	0.45	292	F	2	64–59[f]	*
Sw1119	7	3	0.57	294	B	1.5	60	***
Average	5.60	3.53	0.49	228	-	-	-	-

[a] Number of alleles described in the references or found on the parental animals of RN pedigree.
[b] PIC calculated on average on 20 *RN* parental animals.
[c] Number of informative meioses analyzed.
[d] References: A, this article; B, Rohrer et al. 1994; C, Ellegren et al. 1993; D, Archibald et al. 1995; E, Wilke et al. 1994; F, C. Beattie, personal communication.
[e] Quality: (***) Best markers, automatic interpretation grade; (**) good markers, but several verifications of the automatic interpretation results may be required; (*) usable, but not very efficient or needs careful interpretation.
[f] Touch Down PCR protocol (see Materials and methods).

Genetic distance and linkage desequilibrium calculation. Average and sex-specific, two-point genetic distance estimates were performed with GEMMA. Transmission of both alleles of each parent was also verified with GEMMA. When only one allele was transmitted, the genotype was verified and, if necessary, corrected. Then multiple-point linkage analysis was performed with CRIMAP 2.4 (Green et al. 1990). Order of markers was verified using flips3 and flips4 options on sets of ordered markers. The *RN* gene was localized on the map using the build option. Lower and upper limits of confidence interval for recombination fractions were calculated as the θ value for which $LOD(\theta) = LOD_{max}{}^{-1}$. Existence of linkage disequilibrium was tested on parental haplotypes with the Fisher's Exact Test using Splus statistical package.

Metaphase preparations and chromosome banding. Metaphase slides were prepared from cultured lymphocytes of a normal male pig according to standard procedures. Slides were G-banded with heat denaturation followed by Wright staining, before DISC-PCR experiments. Briefly, slides were incubated in 2× SSC buffer for 2–3 s at 45°C, then stained for 2 min in a mix composed of 1 volume of Wright (Sigma) at 0.25% in methanol/3 volumes of Sorensen buffer (pH 6.8). The slides were RNAse treated for 1 h at 37°C (final concentration 100 µg/ml) and then dehydrated in an ethanol series (70%, 80%, 90%), followed by air drying.

DISC-PCR. DISC-PCR experiments were performed according to Troyer (Troyer et al. 1994) with some modifications. Briefly, the reaction mixture contained, in 40 µl, 150 ng of each oligonucleotide primer; 0.2 mM each dATP, dCTP, dGTP; 0.1 mM Bio16-dUTP or Bio11-dUTP (Boehringer Mannheim); 0.1 mM dTTP; 1.5 mM $MgCl_2$; 0.01% bovine serum albumin (BSA); 1× Taq DNA polymerase buffer; and 5U of Taq DNA polymerase (Appligene or Perkin Elmer). Amplifications were performed in an OmniGene thermal cycler (Hybaid). The thermal profile was for S0088: 94°C for 5 min, 55°C for 1 min, 72°C for 1 min, followed by 23 cycles of 94°C for 1 min, 55°C for 1 min, 72°C for 1 min, and a last cycle of 94°C for 1 min, 55°C for 1 min, 72°C for 5 min. For SW936, it was slightly different: 94°C for 5 min, 57°C for 2 min, 72°C for 2 min, followed by 23 cycles of 94°C for 1 min, 57°C for 2 min, 72°C for 2 min, and at last a cycle of 94°C for 1 min, 57°C for 2 min, 72°C for 8 min.

Signal detection. The slides were allowed to cool to room temperature; the enamel, rubber cement, and cover slide were removed. The slides were immediately immersed in 4× SSC for 5 min and then in blocking buffer composed of 100 mM Tris/HCl (pH 7.5), 150 mM NaCl, 0.5% blocking reagent (Boehringer Mannheim) for 1 h at room temperature. Each slide was then treated with streptavidin-horseradish peroxidase (Vector, Biosys) at a concentration of 5 µg/ml in detection buffer (1× PBS, 1% BSA, and 5 mM EDTA) for 1 h at 37°C in a humidity chamber. The slides were then washed twice for 10 min in 1× PBS at room temperature. They were blocked with 5% goat serum prepared in buffer detection for 5 min at room temperature and then treated with biotinylated anti-streptavidin (Vector, Biosys, France) at a concentration of 2.5 µg/ml in goat serum/detection buffer for 1 h at 37°C. Washing was done as previously, and another treatment with streptavidin-horseradish peroxidase was done as above. Amplification signals were detected by the diaminobenzidine-hydrogen peroxyde reaction (Kit Biosys, France). Slides were restained in 10% Giemsa solution in Sorensen buffer for at least 20 min. The G-banded metaphases photographed before the PCR amplification were relocated and photographed again. Chromosomes bearing signals were identified.

Results

Constitution of a panel of markers of Chr 15. In order to develop a map of Chr 15 in the *RN* pedigree, we tested 24 markers: 20 markers previously assigned to Chr 15 (Archibald et al. 1995; Rohrer et al. 1994), 2 new markers from our laboratory, marker *Sw1119* from the previously unassigned group V (Rohrer et al. 1994), and the unpublished marker *Sw604* (C. Beattie, personal communication of primer sequences). No amplification has been obtained with markers *S0004, Sw1204, Sw159,* and *Sw15.* No interpretable patterns on the ABI sequencer have been obtained for *S0284.* The polymorphism of the retained markers was checked on 10 parents heterozygous for *RN.* All markers except markers *Sw184* and *Sw938* were found to be polymorphic on *RN* heterozygous parents (Table 1). A grade has been given to each marker, from * to ***: *** for the best markers usable in automatic interpretation without special attention; ** for good markers for which some verifications of the automatic interpretation results might be required; * for markers usable, but not very efficient or needing careful interpretation.

Fine mapping of RN gene. A map was produced with 16 markers informative in our *RN* pedigree (Fig. 1). A first network has been established with markers *Sw1111, S0088, Sw1065, Sw1263, Swr1002, Sw120, Sw936, Sw604,* and *Sw1119* by use of CRIMAP.

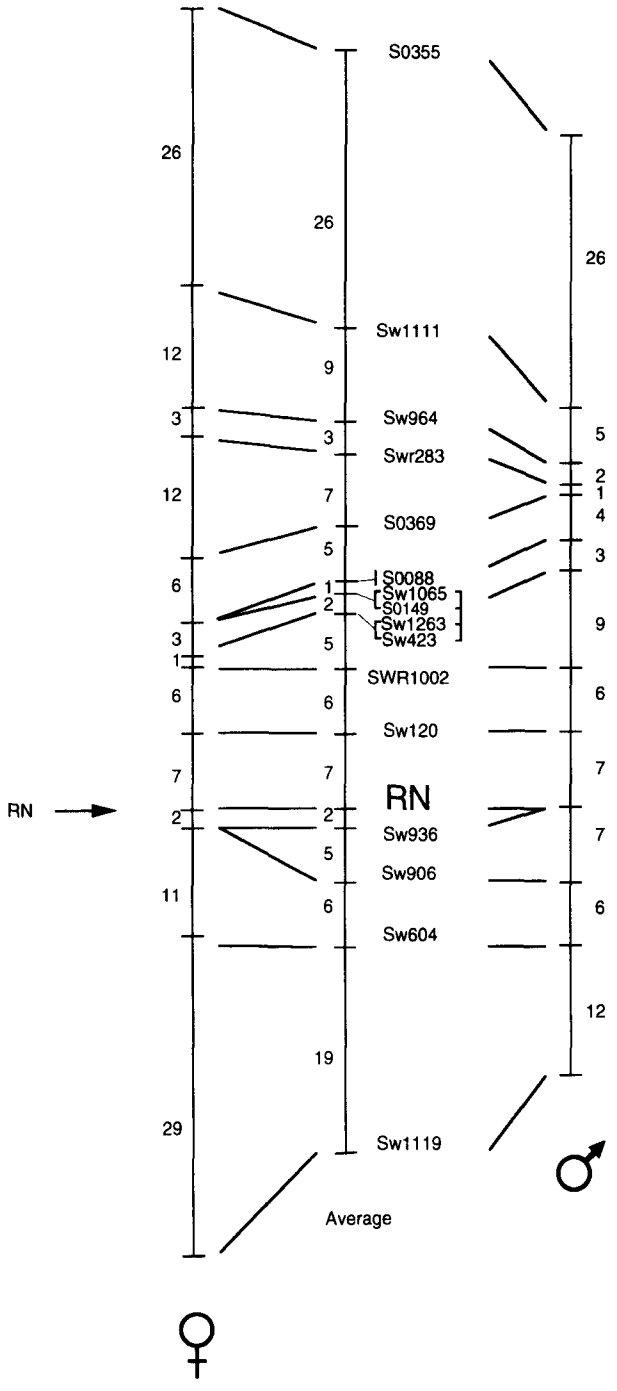


Fig. 1. Multipoint map of pig Chr 15 and mapping of *RN* gene. Distances between adjacent loci are expressed in Kosambi map unit.

Removal or addition of one of these markers did not affect the length of the map. The other markers have been added on this first map. *RN* gene was then put on the map. The analysis of segregation of its alleles allowed us to position RN between markers *Sw120* and *Sw936,* at 1.9 cM from Sw936 with a LOD value of 38.1 and a confidence interval of 0.4–5.2 cM (Table 2). A significant gametic association was observed between the alleles of *RN* and of the flanking markers *Sw120* ($P = 3.10^{-5}$) and *Sw936* ($P = 10^{-7}$) alleles. The gametic association was also significant between *Sw120–Sw936* haplotypes and *RN* alleles ($P = 9.10^{-8}$). The haplotype 2-2 for *Sw120–Sw936* was the most frequent haplotype (28/52) in the 26 parental animals (Table 3). It was always found associated with allele rn^+ (28/28). The haplotype 1-1 (8/52) was frequently found (7/8) associated with RN^- allele.

Localization of RN flanking markers by DISC-PCR. To localize *RN* gene on the physical map of Chr 15, we localized markers *S0088* and *Sw936* by DISC-PCR. Two-hundred-forty (respectively 160) metaphases have been analyzed for *S0088* and *Sw936* respectively. In both experiments, around 50% of analyzed metaphases were G-banded. A hybridization signal was observed on only 10–20% of metaphases, but double spots were frequently observed, especially for *S0088. S0088* was assigned to 15q21, and *Sw936* to the limit between 15q21 and 15q22 (Fig. 2).

Discussion

Establishment of a map of Chr 15. In a previous study, we have found that the *RN* gene is located on Chr 15, at a two-point distance of 18 cM from *S0088* (LOD = 10.1; Milan et al. 1995). To map the *RN* gene more precisely, we analyzed previously published markers of Chr 15 on our pedigree. Out of 26 tested markers, we were able to analyze successfully 18 markers on an ABI sequencer. Sixteen markers were found to be informative on our pedigree, the average PIC value being 0.49. Considering that the experimental families were constructed from only 15 animals belonging to Laconie synthetic line, we were surprised by this relatively high heterozygosity. It gives indications that the markers developed, selected, and mapped on experimental families can also be satisfyingly informative on commercial selected animals. After identification of QTLs on experimental crossing between very different breeds, we will have to determine whether these QTLs have similar effects on commercial lines. Such a synthetic line presenting a high heterozygosity at marker level might be particularly interesting to answer this question.

We produced a map of Chr 15 on *RN* animals. We found the same order as Rohrer and associates on the USDA map (Rohrer et al. 1994). Comparable genetic distances between markers were also observed. The total length of our map is 103 cM, whereas the USDA map on WWW server presents a 132 cM map. However, as

Table 2. Two-point distances between *RN* and flanking markers.

Interval	Sex	No. meioses		Recombination rate (%)	LOD score	Confidence interval[b]
		Recombinant[a]	Informative			
Sw120–RN	♂[c]	3	37	8.1	5.7	(1.7–21.2)
	♀	7	96	7.3	16.5	(2.9–14.3)
	♂ & ♀	10	133	7.5	22.2	(3.7–13.3)
RN–Sw936	♂	0	31	0	8.7	(0–7.2)
	♀	3	126	2.4	29.7	(0.5–6.5)
	♂ & ♀	3	157	1.9	38.1	(0.4–5.2)
Sw120–Sw936	♂	0	8	0	2.1	(0–25.0)
	♀	9	91	9.9	12.8	(4.5–18.0)
	♂ & ♀	9	99	9.1	14.6	(4.1–13.6)

[a] Most probable phase assumed.
[b] The limits of the confidence interval were calculated as the θ values for which $LOD(\theta) = LOD_{max} - 1$.
[c] ♂, male; ♀, female.

Table 3. Haplotype analysis of RN and flanking markers in the 26 parental animals of the pedigree.

	Sw120–Sw936 haplotypes[a]							
	1 – 1	2 – 2	1 – 2	2 – 1	3 – 2	3 – 1	5 – 2	5 – 1
rn^+	1	28	3	3	2	0	1	1
RN^-	7	0	0	1	0	2	1	2
Fisher exact test: $P = 9.10^{-8}$[b]								

[a] The observed size of the *Sw120* alleles were: 1, 156 bp; 2, 145 bp; 3, 160 bp; 5, 149 bp. The observed size of the *Sw936* alleles were: 1, 98 bp; 2, 111 bp.
[b] The *P* value of Fisher exact test indicates the probability that allelic associations are randomly distributed.

we had some information about the location of *RN* (Milan et al. 1995), we did not analyze markers located near the centromere of Chr 15.

Mapping of the RN gene. *RN* gene was mapped between markers *Sw120* and *Sw936,* at 2 cM from *Sw936* (LOD = 38.1). Eight of 204 *RN* genotypes implied double recombinations between *Sw120* and *RN* and between *RN* and *Sw936.* As the two flanking markers are 10 cM apart, and as *RN* genotypes were determined with a probability of 0.9, we eliminated *RN* genotypes implying double recombination. Then, around 95% of genotypes of backcross animals were correct, which is in agreement with the estimated reliability of these genotypes. After suppression of double recombinants, we observed a total of 13 recombinations in the region *Sw120–RN–Sw936.* It is very probable that all *RN* genotypes are correct in this interval. Furthermore, after removing double recombinant meioses, addition of *RN* to the map between *Sw120* and *Sw936* did not increase the distance between these two markers.

In the chromosomal region where *RN* gene has been found, no difference between sexes has been observed. In our previous paper (Milan et al. 1995) we presented a two-point distance between *RN* and *S0088* of only 7 cM during male meiosis, whereas we present here a multipoint distance of 25 cM. This difference might be due only to the limited number of meioses analyzed previously, but it also might be due to a genetic reason, as the two informative males are half-sibs, and as they present the same two haplotypes at markers *S0088–Sw1065–RN–Sw936.* For clarification of this point, genetic distances between *RN* and flanking markers should be analyzed in other populations carrying the RN^- allele. On the map presented here, the multipoint distance of 33 cM observed between *S0088* and *Sw936* is in good agreement with the 30-cM multipoint distance previously published between those markers (Rohrer et al. 1994).

The location of *RN* in the *Sw120–Sw936* chromosomal region was reinforced by the significant allelic association found between *RN* and the flanking markers *Sw120* and *Sw936.* On 52 analyzed haplotypes, the haplotype 2-2 for *Sw120–Sw936* was found associated with the rn^+ allele. In our experimental families, it could thus be selected in order to increase the rn^+ allele frequency. Similarly, haplotype 1-1 should be avoided, as it was frequently found associated with RN^- allele. It remains to be determined whether this allelic association is due only to a founder effect in our experimental pedigree or whether a linkage disequilibrium could also be observed in the Hampshire breed. In any case, it is very likely that a similar disequilibrium exists in other synthetic lines harboring Hampshire blood.

Regional assignment of RN gene on Chr 15. Two microsatellites were mapped on Chr 15 with DISC-PCR (Troyer et al. 1994). The lengths of the amplified DNA fragments were only 100 and 150 bp. As an improvement of the published technique, we were able to realize chromosomal G-banding before amplification, in order to identify the chromosomes. DISC-PCR up to now is not very efficient, because only 10–20% of metaphases were labeled, but it allows a regional mapping of short DNA fragments. For each marker, the labeling of Chr 15 was very similar on the labeled metaphases, and a precise location has been determined. We are confident in the reliability of the localizations that we obtained with this technique: *S0088,* which was mapped at 3 cM from DPP4 (Ellegren et al. 1994), was assigned to 15q2.1 as DPP4 (Thomsen et al. 1993).

The mapping of *S0088* and *Sw936* (this article) and of *DPP4* and *LCT* (Ellegren et al. 1994) on both genetic and cytogenetic maps orientates the Chr 15 linkage group. The limited number of observed labeled metaphases leads us to be cautious about the precise location of *RN* gene, which should be found within the 15q2.1–2.3 region. This localization has to be confirmed by the mapping of other markers flanking the *RN* gene, and eventually by the mapping of *RN* gene itself.

Potential candidate genes for the RN gene. The pig Chr 15 is homologous to human Chr 2q (Rettenberger et al. 1995; Gowreau, pers. comm.), and the 15q2.1–2.3 region is likely to correspond to 2q23–31 region (Lahbib-Mansais et al. 1995). It is, therefore, possible to search in this region for known genes involved in glyco-

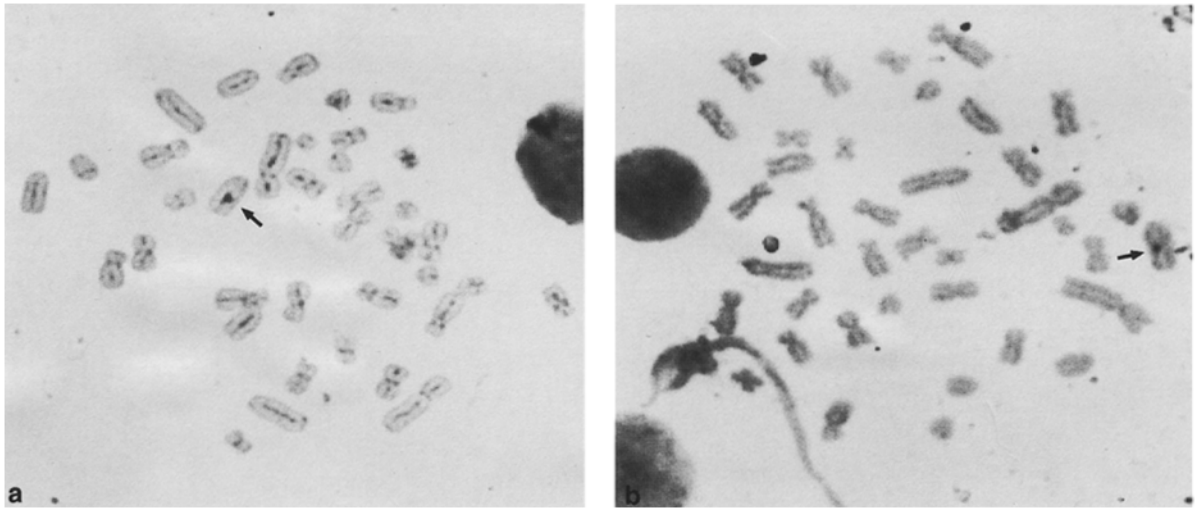


Fig. 2. Localization of **(a)** *S0088* at 15q2.1 and **(b)** *Sw936* at 15q2.1–q2.2 on porcine chromosomes by DISC-PCR. The arrows indicate hybridization signals.

lytic or glycogenic pathways. As an example, on human genome the glucagon gene is located at 2q23, and uridyl transferase 2 gene is assigned to Chr 2. We will search for polymorphism in such genes in order to determine by linkage studies whether they might be "the" *RN* gene, or whether they might participate in an *RN* multigenic locus. Alternatively, we also intend to test several good candidate genes identified by the physiological approach (Estrade et al. 1994), even if they are not located on porcine Chr 15 or its human equivalent. As an example, glycogen branching enzyme 1 (GBE1) is located on human Chr 3 (Thon et al. 1993), which corresponds to pig Chr 13 (Rettenberger et al. 1995). Thus, GBE1 gene itself could not be involved in the RTN trait, but a putative GBE2 gene located on Chr 15 might be responsible for this trait. The availability of porcine somatic hybrid panels (Rettenberger et al. 1994; Yerle et al. unpublished results) and the improvement of the efficiency of DISC-PCR technology should allow a rapid screening of such candidate genes. These two strategies are both envisaged to try to identify candidates that might be the *RN* gene.

Acknowledgments. We thank Magali San Cristobal-Gaudy for her help in statistical tests and Philippe Mulsant, Claude Chevalet, and Alain Vignal for fruitful discussions and critical reading of the manuscript. The authors are very grateful to Craig Beattie for communication of primer sequences prior to publication. This work was supported in part by grants from INRA Agrobio programme, from GREG (Groupement de Recherche et d'Etude des Génomes), and from EU PiGMaP BRIDGE project (DGXII).

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